Shape is Universal

Image-Based Scagnostics Pipeline

1. Pipeline Overview

This pipeline enables the computation of scagnostics (scatterplot diagnostics) from any spatial data representation through image-based intermediaries. The approach generalizes scagnostics beyond traditional point-cloud data to heatmaps, heightmaps, density maps, and other continuous spatial representations.

2. Input Stage

2.1 Accepted Input Types

- Binary grids (0/1 matrices)
- Float point arrays (continuous values)
- Heatmaps and density maps
- Heightmaps and elevation data
- Any rasterizable spatial data

2.2 Normalization

All inputs are normalized to a standard resolution (recommended: 256×256). This ensures consistent computation across different data sources while balancing detail preservation with computational efficiency.

- Intensity values scaled to [0, 1]
- Spatial dimensions resampled to target resolution
- Sensitivity testing recommended across resolutions (128, 256, 512)

3. Preprocessing Stage

3.1 Adaptive Gaussian Smoothing

Apply Gaussian blur to unify data representation. The smoothing kernel should be adaptive based on input type:

- Binary inputs: σ = 2-4 pixels (effectively KDE with Gaussian kernel)
- Already-smooth inputs (heatmaps): σ = 0.5-1 pixel or skip
- Default recommendation: σ = resolution / 128 (e.g., σ = 2 for 256×256)

4. Shape Extraction Stage

4.1 Contour Extraction (Alpha Shape Proxy)

Use Marching Squares algorithm to extract isocontours at multiple levels, serving as a continuous analog to traditional alpha shapes.

- **Primary contour:** 10th percentile of non-zero values (outer boundary)
- **Secondary contours:** 25th, 50th, 75th percentiles (internal structure)
- Alpha area = area enclosed by primary contour

4.2 Convex Hull Computation

Compute the convex hull from the outermost contour points using standard algorithms (e.g., Graham scan, Quickhull).

- Input: Points from the primary (outermost) contour
- Output: Convex hull area for shape metrics

5. Graph Extraction Stage (MST Equivalent)

Extract a graph structure that captures the connectivity and branching of the data distribution. The recommended approach preserves continuous intensity information.

5.1 Primary Method: Gradient-Based Graph Construction

This method respects the continuous nature of the data by following the natural ridges of the density field:

- **Step 1 -** Compute gradient field: $\nabla I(x,y)$ from the smoothed image
- **Step 2 -** Identify critical points: Local maxima become graph nodes
- **Step 3 -** Trace ridge lines: Connect nodes via gradient descent paths
- **Step 4 -** Extract edge lengths: Geodesic distances along ridges

5.2 Alternative Methods

Option A: Threshold + Skeletonization

- Threshold at T (e.g., 0.2 or adaptive Otsu threshold)
- Apply morphological skeletonization (medial axis transform)
- Convert skeleton to graph: junctions → nodes, paths → edges

Option B: Discrete Curve Evolution

- Apply skeleton pruning via DCE algorithm
- Robust to noise; principled pruning of spurious branches
- Best for shapes with clear boundaries

6. Density Metrics Extraction Stage

This stage extracts the statistical properties needed to compute density-based scagnostics (Skewed, Clumpy, Striated, Outlying, etc.).

6.1 Intensity Distribution Analysis

Analyze the histogram of pixel intensities within the alpha contour:

- **Skewness:** Third standardized moment of intensity distribution
- **Kurtosis:** Fourth standardized moment (peakedness/tail weight)
- **Entropy:** Shannon entropy of intensity histogram

6.2 Local Variance Analysis (Clumpiness)

Compute local variance to detect clustering and non-uniformity:

- **Method:** Sliding window variance (window size: 16×16 pixels)
- **Clumpiness Index:** Coefficient of variation of local variance map
- High values indicate clustered/clumpy distributions

6.3 Structure Tensor Analysis (Striation)

Detect directional patterns using the structure tensor:

- **Compute:** Structure tensor $S = [I_x^2, I_xI_y; I_xI_y, I_y^2]$ averaged over windows
- **Extract:** Eigenvalues λ_1, λ_2 of the structure tensor
- **Striation Index:** $(\lambda_1 - \lambda_2) / (\lambda_1 + \lambda_2)$ — coherence measure
- Values near 1 indicate strong directional patterns

6.4 Persistence Analysis (Outlying)

Use topological methods to identify outliers and isolated components:

- **Method:** Threshold sweep from max to min intensity
- **Track:** Connected component count at each threshold (Betti number β_0)
- **Persistence:** Lifetime of components (birth - death threshold)
- **Outlying Index:** Ratio of short-lived to total components

7. Scagnostic Measures Computation

Map extracted features to standard scagnostic measures:

Measure	Formula / Method	Image-Based Source
Outlying	length(longest MST edge) / length(q90 edge)	Persistence ratio or isolated component count
Skewed	Skewness of MST edge lengths	Intensity histogram skewness
Clumpy	RUNT statistic on MST	Local variance coefficient of variation

Sparse	q90(MST edge length)	Mean distance between contour levels
Striated	Angle coherence in MST	Structure tensor coherence
Convex	Area(alpha hull) / Area(convex hull)	Area(primary contour) / Area(convex hull)
Skinny	1 - √(4π × Area) / Perimeter	Same formula on primary contour
Stringy	Diameter(MST) / length(MST)	Graph diameter / total edge length
Monotonic	Spearman correlation²	Principal axis alignment of intensity

8. Complete Pipeline Flow

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┌──────────────────────────────────────────────────────────────┐
│                        INPUT STAGE                           │
│   Binary Grid / Float Points / Heatmap / Density             │
│                            ↓                                 │
│            Normalize to 256×256, scale to [0,1]              │
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                               ↓
┌──────────────────────────────────────────────────────────────┐
│                     PREPROCESSING STAGE                      │
│          Adaptive Gaussian Blur (σ = resolution/128)         │
└──────────────────────────────────────────────────────────────┘
                               ↓
              ┌────────────────┼────────────────┐
              ↓                ↓                ↓
     ┌─────────────┐   ┌─────────────┐   ┌──────────────────┐
     │   SHAPE     │   │   GRAPH     │   │     DENSITY      │
     │ EXTRACTION  │   │ EXTRACTION  │   │     METRICS      │
     ├─────────────┤   ├─────────────┤   ├──────────────────┤
     │ Marching Sq.│   │ Gradient    │   │ Intensity Histogram│
     │ Contours    │   │ Field +     │   │ Local Variance   │
     │ Convex Hull │   │ Critical Pts│   │ Structure Tensor │
     │             │   │ Ridge Lines │   │ Persistence (H0) │
     └─────────────┘   └─────────────┘   └──────────────────┘
              │                │                │
              ↓                ↓                ↓
     ┌─────────────┐   ┌─────────────┐   ┌──────────────────┐
     │ Alpha Area  │   │ Edge Lengths│   │ Skewness, Kurtosis│
     │ Convex Area │   │ Graph Diam. │   │ Clumpiness Index │
     │ Perimeter   │   │ Branch Count│   │ Striation Index  │
     │             │   │             │   │ Outlying Ratio   │
     └─────────────┘   └─────────────┘   └──────────────────┘
              └────────────────┼────────────────┘
                               ↓
┌──────────────────────────────────────────────────────────────┐
│                    SCAGNOSTIC COMPUTATION                    │
```

```
Outlying | Skewed | Clumpy | Sparse | Striated | Convex |
Skinny | Stringy | Monotonic
```

9. Implementation Notes

9.1 Recommended Libraries

- **Python:** scikit-image (contours, skeletonization), scipy (structure tensor), GUDHI or Ripser (persistence)
- **C++:** OpenCV (image processing), CGAL (computational geometry), Dionysus (persistence)
- **JavaScript:** D3.js (contours), ml-matrix (linear algebra)

9.2 Hyperparameters to Document

- Resolution: 256×256 (default)
- Gaussian σ: resolution / 128
- Contour isovalues: 10th, 25th, 50th, 75th percentiles
- Local variance window: 16×16 pixels
- Threshold for skeleton (if used): Otsu or 0.2

9.3 Validation Strategy

To validate the “Shape is Universal” claim:

- Generate synthetic point clouds with known scagnostic properties
- Compute traditional scagnostics on point clouds
- Render point clouds as images and compute image-based scagnostics
- Verify strong correlation ($r > 0.9$) between methods
- Test sensitivity to resolution, blur, and threshold parameters

10. Edge Cases and Limitations

- **Sparse data:** May require larger blur σ or adaptive bandwidth selection
- **Saturated regions:** Clip or normalize to prevent loss of structure
- **Multi-modal distributions:** Persistence analysis handles naturally; may need multiple contour levels
- **Noise sensitivity:** Preprocessing blur mitigates; DCE provides robust pruning